THE EVALUATION MAP ON INTERNAL HOM IS AN ISOMORPHISM FOR PERFECT FIRST ARGUMENT OR FOR PSEUDO-COHERENT FIRST ARGUMENT WITH BOUNDED BELOW SECOND AND FINITE INJECTIVE DIMENSION THIRD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The statement contains the author's lowercase sentence opening 'the map', after a full stop; it is reproduced unchanged. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- K-injective complexes and the derived Hom as a complex of Hom groups. used at proof lines 28578-28588. Supplies the explicit complexes between which the evaluation map is compared.
- Finitely generated projective modules and the adjunction Hom_R(J, I) tensor_R K = Hom_R(Hom_R(K, J), I). used at proof lines 28593-28600. Is the reason each component map c_{p,r,s} is an isomorphism.
- Boundedness and bounded-below conditions on complexes and their effect on the indices occurring in a Hom complex. used at proof lines 28607-28613. Is the case split that makes the finiteness bookkeeping work in each of the two stated situations.

Source reference. Original lemma 18164: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18164–18164. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18732: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18732–18732. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let K, L, M be objects of $D(R)$. the map*

$$R\mathrm{Hom}_R(L, M) \otimes^{\mathbf{L}}_R K \longrightarrow R\mathrm{Hom}_R(R\mathrm{Hom}_R(K, L), M)$$

of Lemma 18732 is an isomorphism in the following two cases

(1) *K perfect, or*
(2) *K is pseudo-coherent, $L \in D^+(R)$, and M finite injective dimension.*

Proof. Choose a K-injective complex $I^\bullet$ representing M, a K-injective complex $J^\bullet$ representing L, and a bounded above complex of finite projective modules $K^\bullet$ representing K. Consider the map of complexes

$$\mathrm{Tot}(\mathrm{Hom}^\bullet(J^\bullet, I^\bullet) \otimes_R K^\bullet) \longrightarrow \mathrm{Hom}^\bullet(\mathrm{Hom}^\bullet(K^\bullet, J^\bullet), I^\bullet)$$

of Lemma 18164. Note that

$$\Big(\prod\nolimits_{p+r=t} \mathrm{Hom}_R(J^{-r}, I^p)\Big) \otimes_R K^s = \prod\nolimits_{p+r=t} \mathrm{Hom}_R(J^{-r}, I^p) \otimes_R K^s$$

because K^s is finite projective. The map is given by the maps

$$c_{p,r,s} : \mathrm{Hom}_R(J^{-r}, I^p) \otimes_R K^s \longrightarrow \mathrm{Hom}_R(\mathrm{Hom}_R(K^s, J^{-r}), I^p)$$

which are isomorphisms as K^s is finite projective. For every element $\alpha = (\alpha^{p,r,s})$ of degree n of the left hand side, there are only finitely many values of s such that $\alpha^{p,r,s}$ is nonzero (for some p, r with $n = p+r+s$). Hence our map is an isomorphism if the same vanishing condition is forced on the elements $\beta = (\beta^{p,r,s})$ of the right hand side. If $K^\bullet$ is a bounded complex of finite projective modules, this is clear. On the other hand, if we can choose $I^\bullet$ bounded and $J^\bullet$ bounded below, then $\beta^{p,r,s}$ is zero for p outside a fixed range, for $s \gg 0$, and for $r \gg 0$. Hence among solutions of $n = p + r + s$ with $\beta^{p,r,s}$ nonzero only a finite number of s values occur. □